

**TECHSPEC® 16mm UC Series Fixed Focal Length Lens**

#33-304



16mm FL, #33-304

Our ultra-compact, TECHSPEC® UC Series Fixed Focal Length Lenses are designed to optimize performance, cost, and size without sacrificing quality or feel. Designed for pixels that are  $\leq 2.2\mu\text{m}$ , these lenses provide high levels of resolution ( $>200\text{ lp/mm}$ ) across the sensor and are compatible with all standard C-Mount cameras. TECHSPEC® UC Series Fixed Focal Length Lenses feature focus and iris adjustments, as well as recessed set screws, and are manufactured for use at typical machine vision working distances. While they are optimized for 1/2.5" sensors, many focal lengths will work on sensors up to 1/1.8". The TECHSPEC® UC Series Fixed Focal Length Lenses are an outstanding option for use on all smaller format camera sensors, along with both short and long working distance applications, making them ideal for inspection, [factory automation](#), biomedical devices, and a broad range of other applications.

**Note:** For fixed aperture instrumentation versions, see our [TECHSPEC® UCi Series Fixed Focal Length Lenses](#).

## General

|                    |                                                |
|--------------------|------------------------------------------------|
| Product Family:    | UC Series                                      |
| Type:              | Fixed Focal Length Lens                        |
| Imaging Lens Type: | High Performance Lens with Compact Form Factor |

## Physical & Mechanical Properties

|                               |          |
|-------------------------------|----------|
| Iris Option:                  | Variable |
| Length (mm):                  | 29.13    |
| Maximum Diameter (mm):        | 30       |
| Outer Diameter (mm):          | 30       |
| Weight (g):                   | 42       |
| Maximum Rear Protrusion (mm): | 3.38     |
| Maximum Length (mm):          | 29.13    |

## Optical Properties

|                                               |                                                                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------|
| Horizontal Field of View @ Max Sensor Format: | 98.5mm - 25.3°                                                                      |
| Field of View at Max Sensor Format:           | Horizontal: 98.5mm - 25.3°<br>Vertical: 73.9mm - 19.1°<br>Diagonal: 123.1mm - 31.3° |
| Horizontal Field of View, 1/1.8" Sensor:      | 98.5mm - 25.3°                                                                      |
| Horizontal Field of View, 1/2" Sensor:        | 87.6mm - 22.5°                                                                      |
| Horizontal Field of View, 1/2.5" Sensor:      | 79.4mm - 20.5°                                                                      |
| Horizontal Field of View, 1/3" Sensor:        | 65.7mm - 17.0°                                                                      |
| Horizontal Field of View, 1/4" Sensor:        | 49.3mm - 12.8°                                                                      |
| Maximum Image Circle (mm):                    | 9.00                                                                                |
| Numerical Aperture NA, Object Side:           | 0.0197                                                                              |
| Number of Elements (Groups):                  | 7 (5)                                                                               |
| Focal Length FL (mm):                         | 16.00                                                                               |
| Working Distance (mm):                        | 200 - ∞                                                                             |
| Aperture (f/#):                               | f/1.8 - f/11                                                                        |
| Coating:                                      | λ/4 MgF <sub>2</sub>                                                                |
| Coating Specification:                        | λ/4 MgF <sub>2</sub>                                                                |
| Entrance Pupil Position (mm):                 | 10.68                                                                               |
| Object Space Principal Plane (mm):            | 20.18                                                                               |
| Image Space Principal Plane (mm):             | -6.67                                                                               |
| Maximum Distortion (%):                       | -0.53                                                                               |
| Exit Pupil Position (mm):                     | -29.94                                                                              |
| Lens Wavelength Range:                        | VIS                                                                                 |

## Sensor

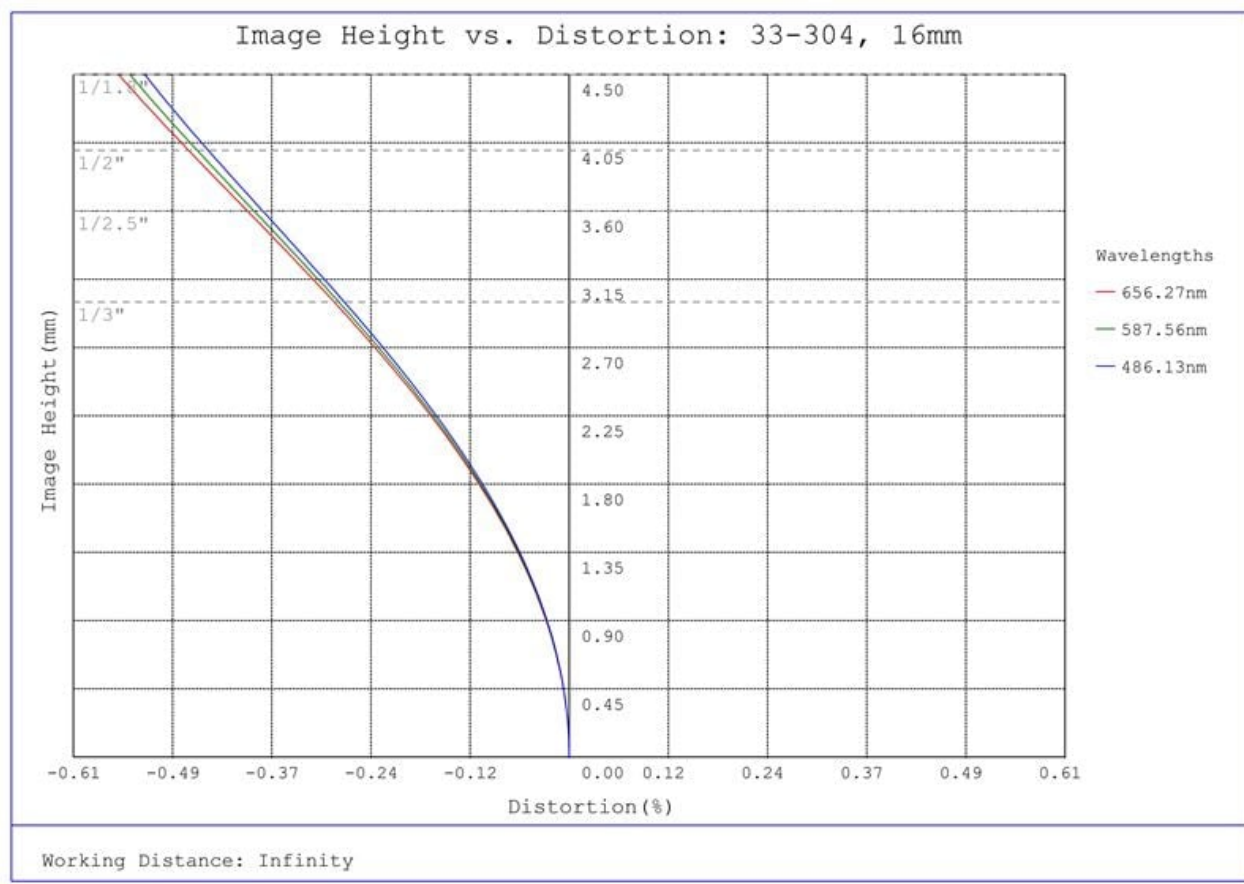
|                          |        |
|--------------------------|--------|
| Optimized Sensor Format: | 1/2.5" |
| Maximum Sensor Format:   | 1/1.8" |
| Pixel Size (μm):         | 1.85   |

## Threading & Mounting

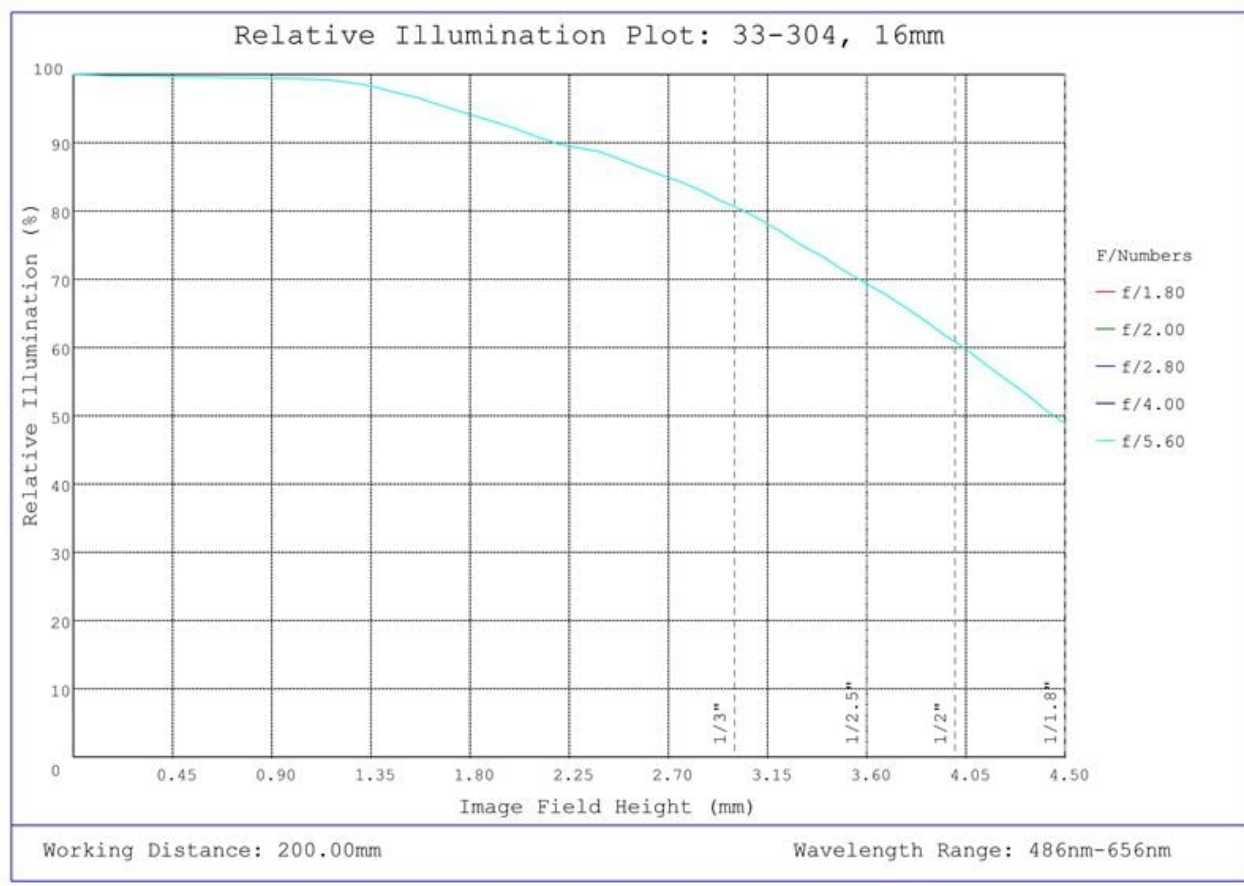
|                         |                       |
|-------------------------|-----------------------|
| Filter Thread:          | M25.5 x 0.50 (Female) |
| Mounting Diameter (mm): | 30.0                  |
| Mount:                  | C-Mount               |

**Environmental & Durability Factors**

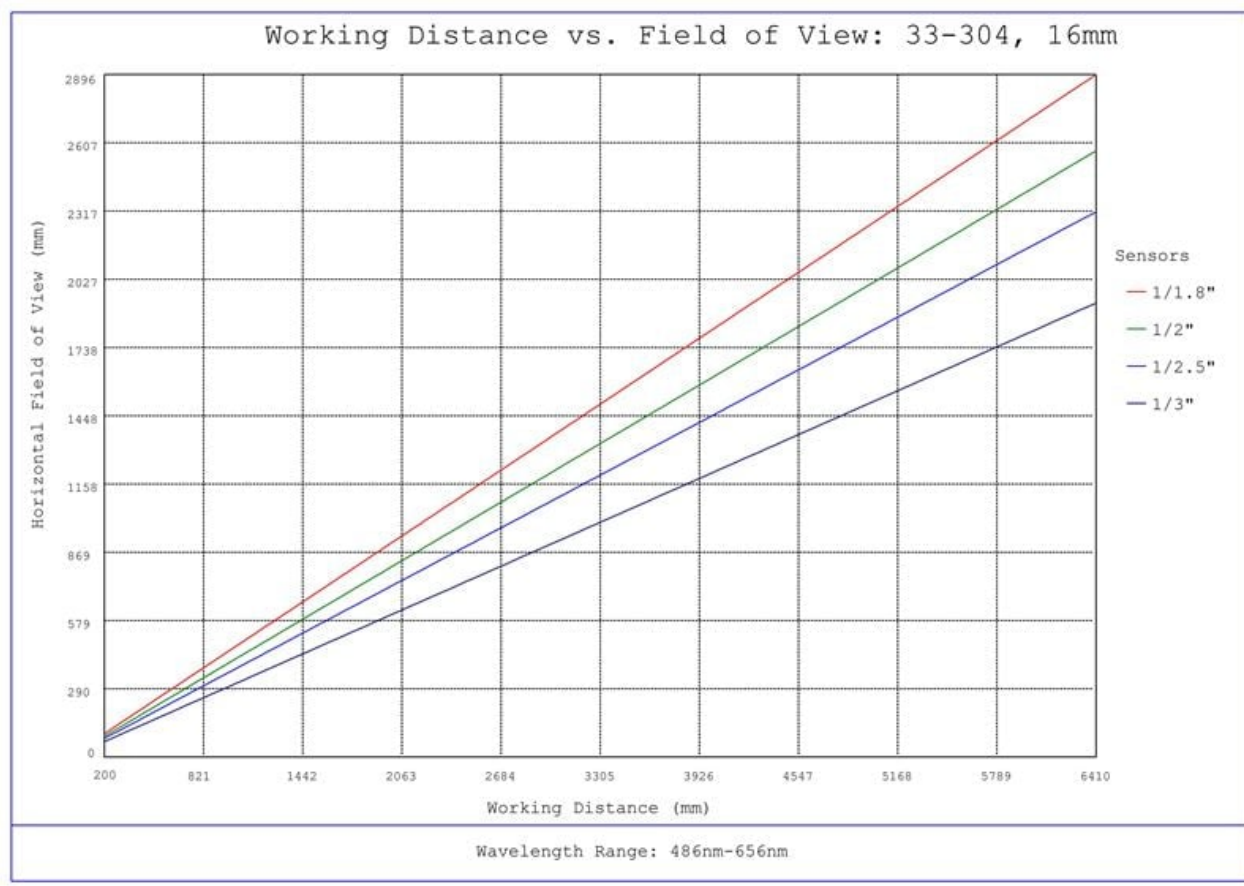
|                           |            |
|---------------------------|------------|
| Storage Temperature (°C): | -20 to +60 |
|---------------------------|------------|



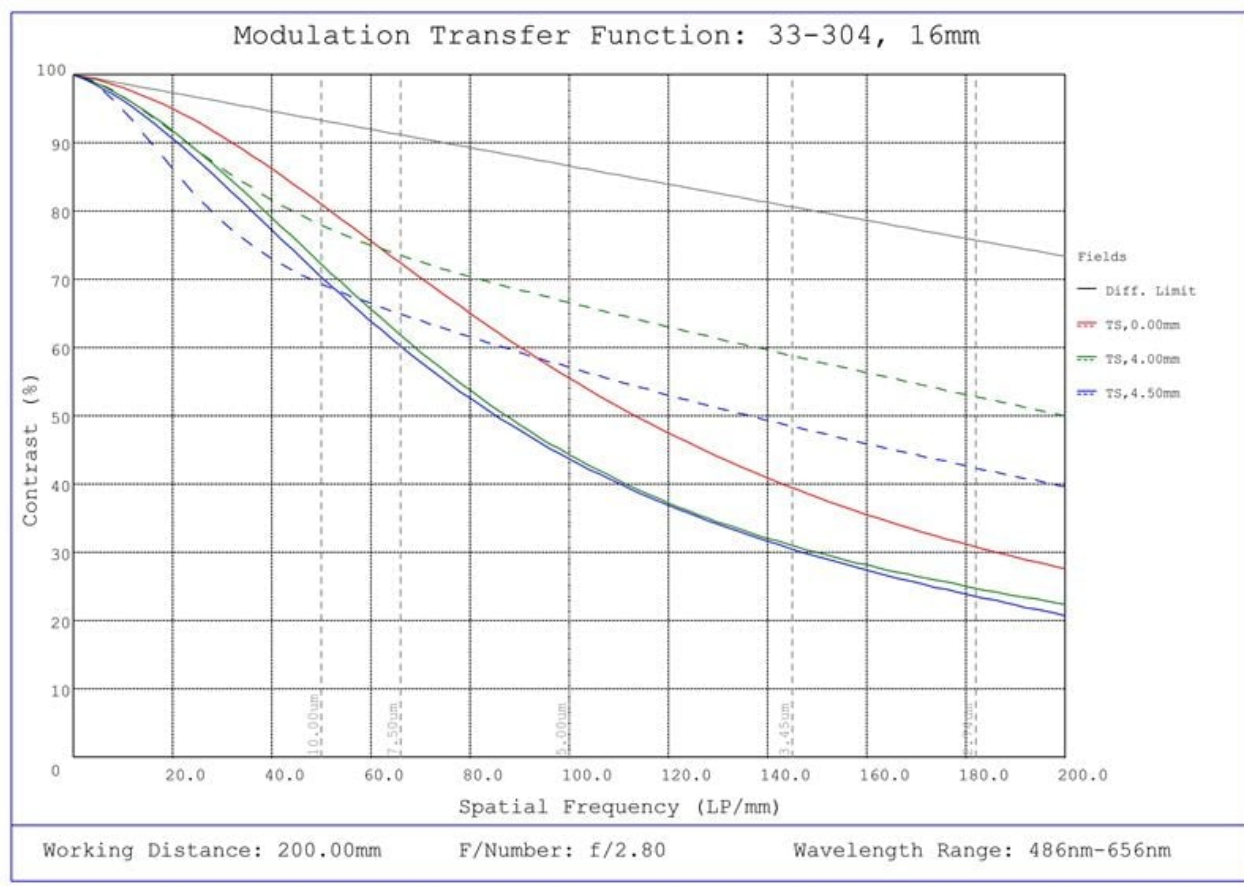
#33-304, 16mm UC Series Fixed Focal Length Lens, Distortion Plot



#33-304, 16mm UC Series Fixed Focal Length Lens, Relative Illumination Plot



#33-304, 16mm UC Series Fixed Focal Length Lens, Working Distance versus Field of View Plot



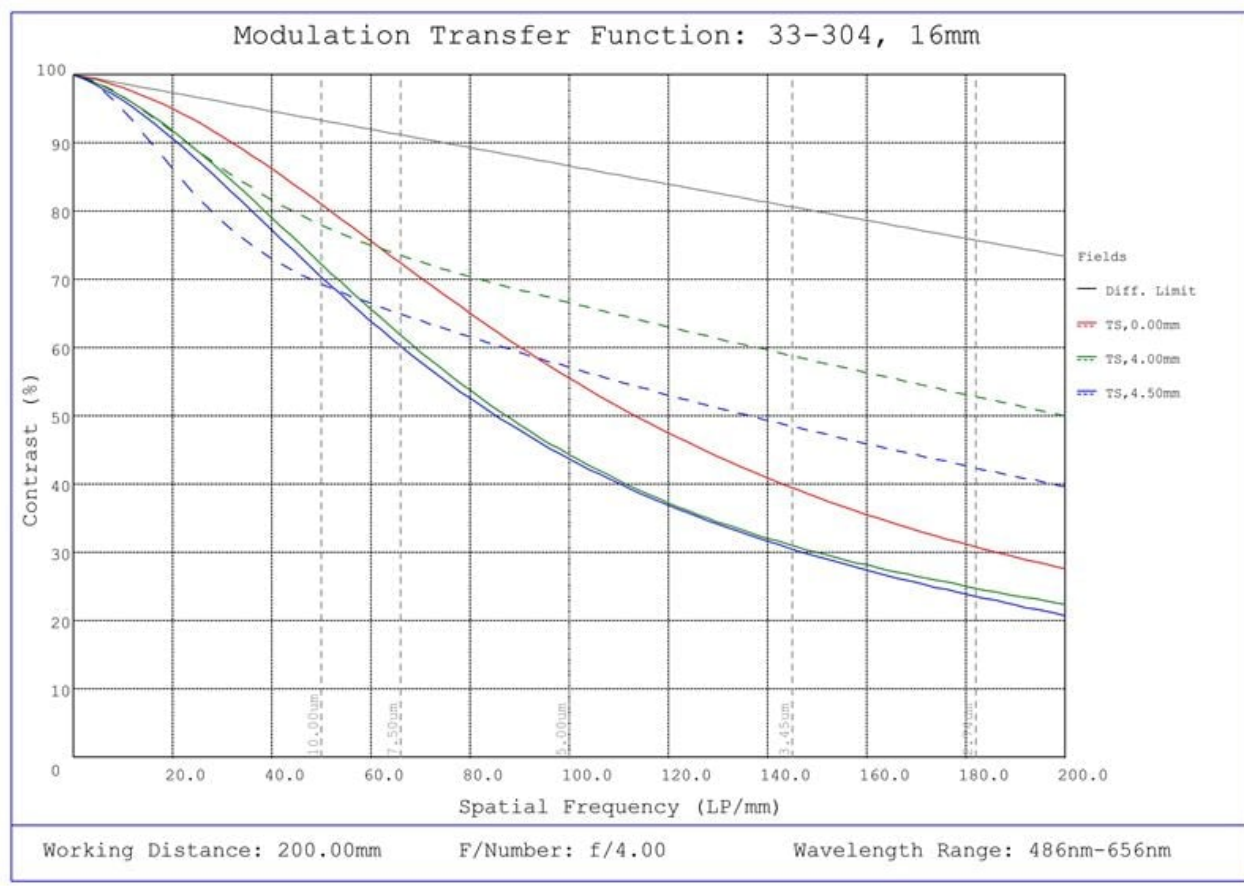
#33-304, 16mm UC Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 200mm Working Distance, f2.8





#33-304, 16mm UC Series Fixed Focal Length Lens, Depth of Field Plot, 200mm Working Distance, f2.8

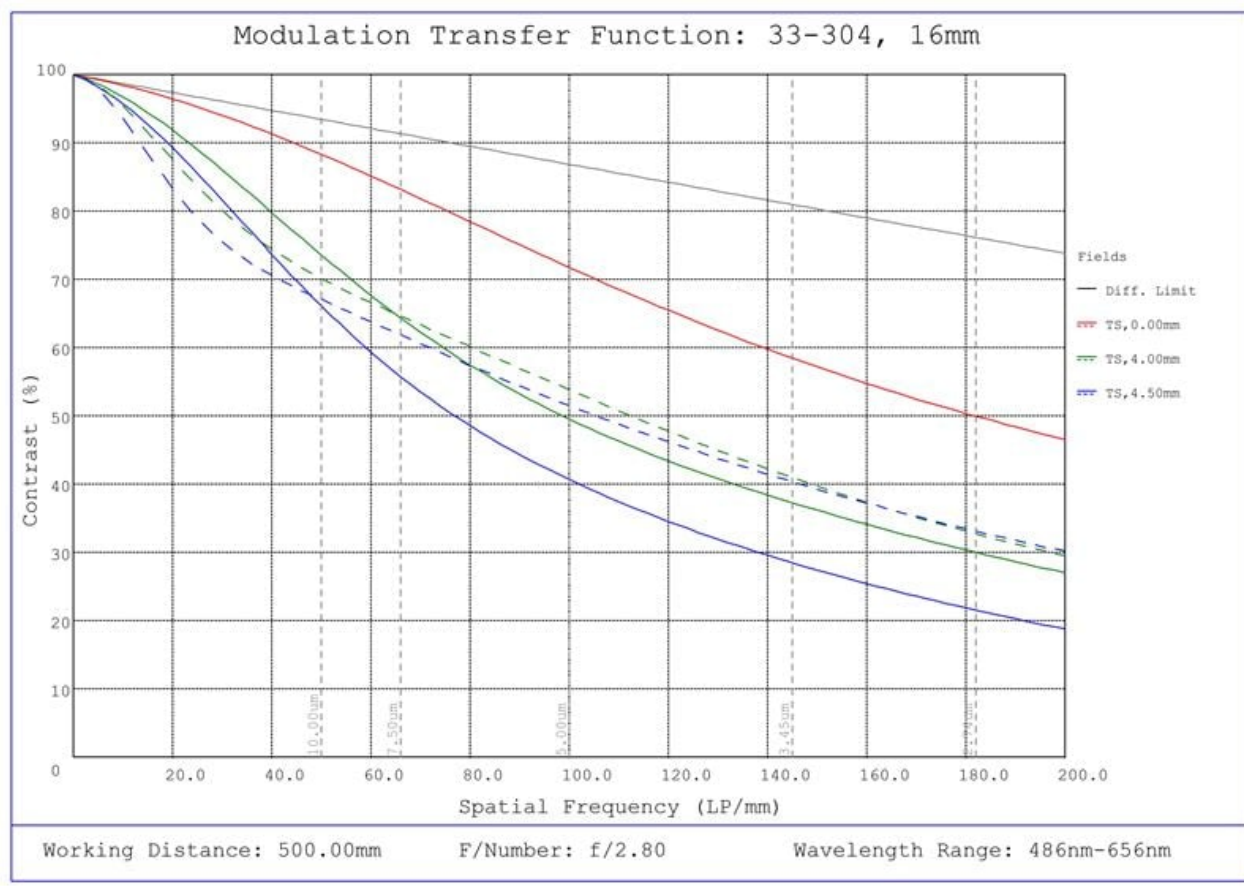




#33-304, 16mm UC Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 200mm Working Distance, f4



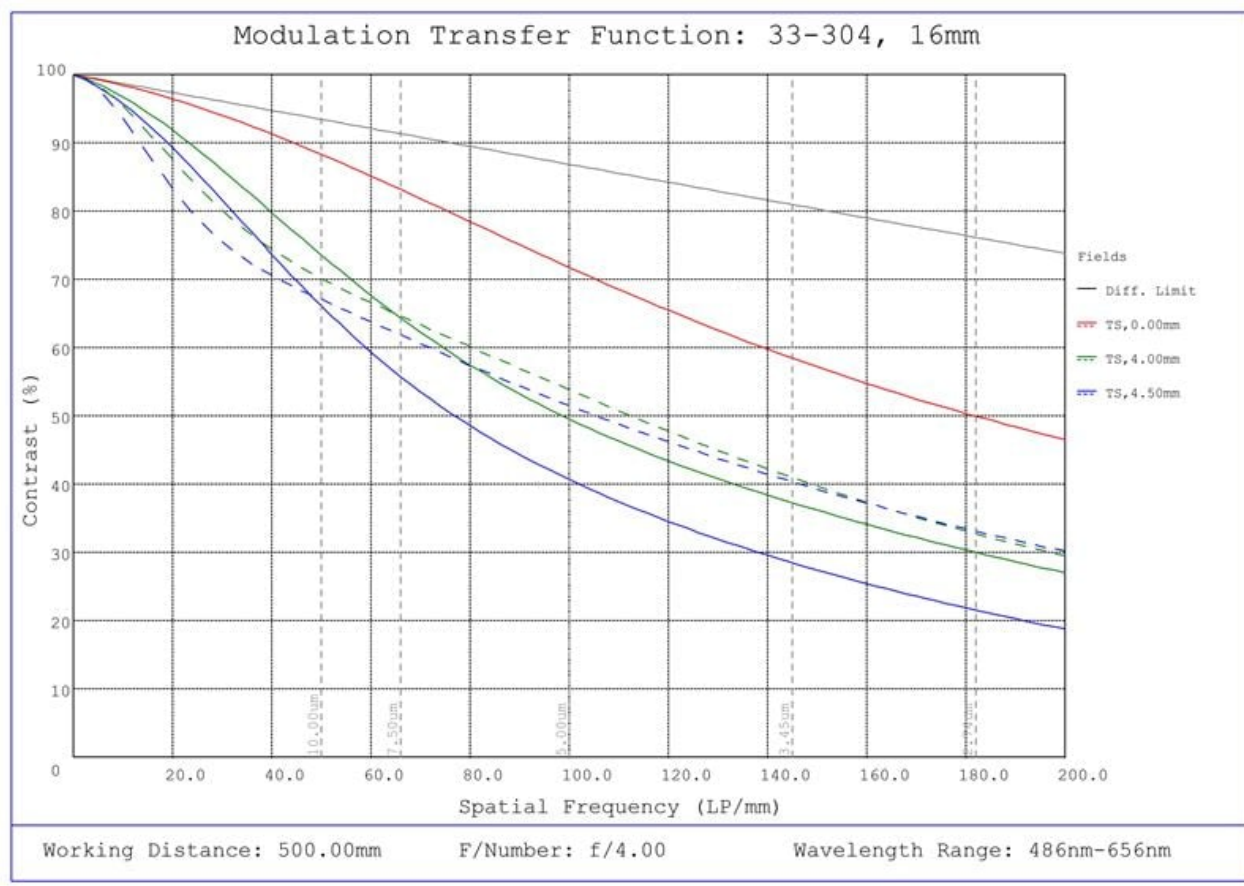
#33-304, 16mm UC Series Fixed Focal Length Lens, Depth of Field Plot, 200mm Working Distance, f4



#33-304, 16mm UC Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 500mm Working Distance, f2.8



#33-304, 16mm UC Series Fixed Focal Length Lens, Depth of Field Plot, 500mm Working Distance, f2.8



#33-304, 16mm UC Series Fixed Focal Length Lens, Modulated Transfer Function (MTF) Plot, 500mm Working Distance, f4





#33-304, 16mm UC Series Fixed Focal Length Lens, Depth of Field Plot, 500mm Working Distance, f4



May 09, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

| Stock No. | Description                            |
|-----------|----------------------------------------|
| 33-304    | 16mm UC Series Fixed Focal Length Lens |

This item is EU RoHS (2015/863/EU) compliant without the use of exemptions.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



Jay Budd, Director of Corporate Compliance  
May 09, 2026

Edmund Optics Inc. - 101 E Gloucester Pike, Barrington, NJ 08007 | 1-800-363-1992 | [Compliance@edmundoptics.com](mailto:Compliance@edmundoptics.com)



May 09, 2026

| Stock No. | Description                            |
|-----------|----------------------------------------|
| 33-304    | 16mm UC Series Fixed Focal Length Lens |

Edmund Optics certifies that all articles included in this shipment are in compliance with the terms and conditions of this order. The company also certifies that the articles included in this shipment are in accordance to all agreed upon specifications and quality assurance provisions. Please call 1-800-363-1992 should any questions arise from this shipment.



Jeff Harvey- EVP of Operations





May 09, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

| Stock No. | Description                            |
|-----------|----------------------------------------|
| 33-304    | 16mm UC Series Fixed Focal Length Lens |

This item does not contain any of the substances listed on the Reach SVHC list as of 25 June 2025 in quantities exceeding 0.1% weight by weight.

Please note the following:

- Entry 11 of Annex V of the REACH regulation EC 1907/2006 as amended by regulation 987/2008, exempts "non-hazardous" glass components from registration.
- Items considered to be packing materials and which do not form part of, or are not required for the operation of, the product, are not considered to be within scope of this declaration.
- EO does not routinely analyze products for substances not purposely added or specified within the design specifications, drawings and assembly procedures.

This certification means that:

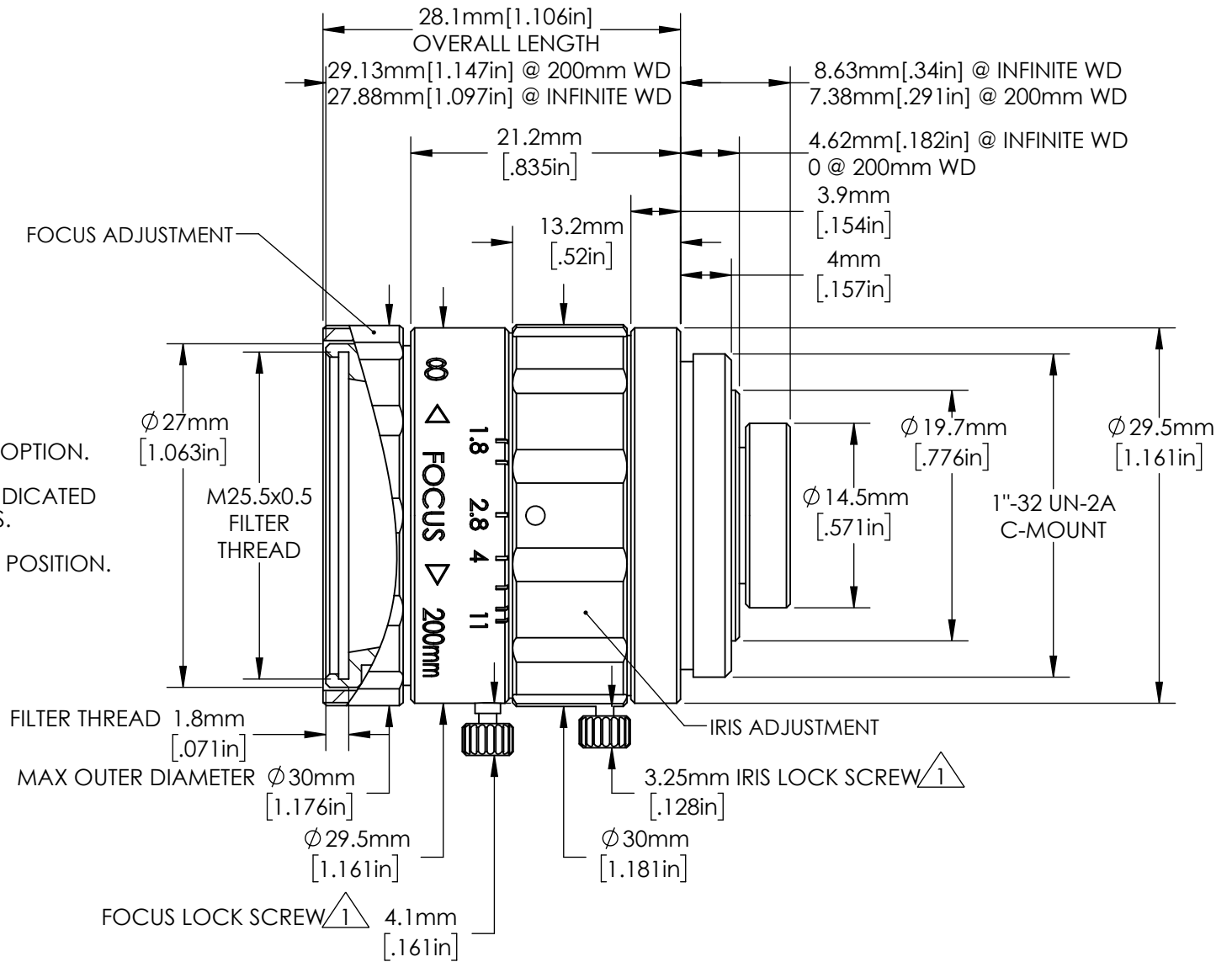
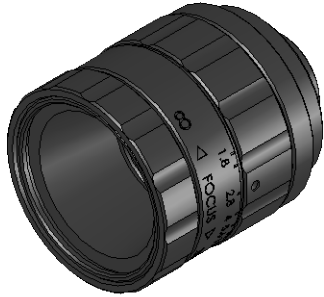
- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



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#### NOTES:

1. INCLUDES RECESSED SET SCREW OPTION.
2. LENS MAY MOVE MORE THAN INDICATED UNTIL IT REACHES BUILT-IN STOPS.
3. LENS SHOWN IN THE INFINITE WD POSITION.

**FOR INFORMATION ONLY:**  
DO NOT MANUFACTURE  
PARTS TO THIS DRAWING

SPECIFICATIONS SUBJECT TO CHANGE  
WITHOUT NOTICE.  
DIMENSIONS ARE FOR REFERENCE ONLY.  
FOR DETAILED SPECIFICATIONS,  
PLEASE REFER TO THE SPEC SHEET.



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|        |                                        |  |              |
|--------|----------------------------------------|--|--------------|
| TITLE  | 16mm UC Series Fixed Focal Length Lens |  |              |
| DWG NO | 33304                                  |  | SHEET 1 OF 1 |

Enable 3D View